MILITARY SPECIFICATION

BOLT, TIBIAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tibial bolt used in orthopedic surgery for repair of fractures of the tibia.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein. These documents may be obtained as directed by the contracting officer.

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metals, Test Methods.

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

10130 BOLT, TIBIAL

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E45-63	Determining the Inclusion Content of Steel.
ASTM F55-66	Stainless Steel Bars And Wire For Surgical Implants.
ASTM F86-68	Recommend Practice for Surface Requirements of Metallic Surgical Implants.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042).

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I. B1.1	Unified Screw Threads.
A.N.S.I. B1.2	Gages and Gaging for Unified Screw Threads.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

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3. REQUIREMENTS

3.1 Material. The tibial bolts shall be fabricated from cold drawn bar or wire conforming to the composition specified in ASTM F55-66 concerning stainless steel material used for surgical implants, except that the carbon content shall be 0.03 percent maximum.

3.1.1 Grain structure. Grain size of all raw material used in the fabrication of the bolts shall be determined and shall be size 6 or finer in accordance with ASTM F55-66.

3.1.2 Inclusion content of material. The nonmetallic inclusion content of all material used in the fabrication of the bolts shall be determined by the JERKONTORET (JK) Chart method in accordance with ASTM E45-63, using method "A" or "D". The numbers listed in Table I shall represent the maximum allowable inclusion ratings.

Table I. Maximum Allowable Inclusion Ratings In Accordance with ASTM E45

	Type A	Type B	Type C	Type D
Thin Series	2	1-1/2	1-1/2	2
Thick Series	1-1/2	1	1	1

3.2 Hardness. Rockwell hardness of all components of the bolts shall be C30 to C35 when tested as specified in 4.4.2.

3.3 Finish. The washers, nuts and heads of the bolts shall have a mirror finish. The bolt proper (excluding washers and nuts) shall not be electropolished. Definitions of acceptable surface characteristics shall be as described in ASTM F86.

3.3.1 Passivation. All components of the tibial bolts shall be properly passivated after completion of all final finishing operations and the passivation shall be accomplished by the methods described in ASTM F86.

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3.4 Style, design and dimensions. Tibial bolts shall be suitable for orthopedic surgery involving fractures of the human tibia. The bolt shall be furnished with 2 mating nuts and 2 mating washers. The bolt proper shall be of one-piece construction. As an alternate, the head of the bolt shall be heliarc welded to the thread member; swaging or other cold working operations shall not be acceptable. The bolt head shall possess a slot, and the distal end of the bolt shall possess a sharp point. The threads of the bolt shall be in accordance with A.N.S.I. Standard B1.1 for 3-48 UNC--2A (Unified Screw Thread System of Dimensions). The threaded portion of the bolt shall extend to and approximate the head of the screw. As an alternate a shoulder of the same diameter of the bolt shall extend to the head of the screw. Tibial bolts shall be in accordance with DPSC Drawing No. 10130.

3.5 External threads of the bolt. External threads of the bolt shall allow the "GO" thread ring gages to pass over their full length and turn freely and shall not allow passage of the "LO" and "NOT GO" thread ring gages when tested as specified in 4.4.3. Definitions of acceptance and rejection of the gaging tests shall be as specified in A.N.S.I. Standard B1.2.

3.6 Internal threads of the nuts. Internal threads of the nuts shall allow the "GO" thread plug gages to be inserted and turned freely, and shall not allow entry of the "HI" and "NOT GO" thread plug gages when tested as specified in 4.4.4. Definitions of acceptance and rejection of the gaging tests shall be as specified in A.N.S.I. Standard B1.2.

3.7 Resistance to boil test, copper sulfate test and saline test. All components of finished tibial bolts shall show no evidence of corrosion when subjected to the boil test as specified in 4.4.1.1, the copper sulfate test as specified in 4.4.1.2, and the saline test as specified in 4.4.1.3.

3.8 Workmanship. Workmanship shall be first class throughout. Screw threads and bolt head slot shall be free from burrs, pits and tool marks. Tibial bolts shall be free from defects which detract from their appearance or impair their serviceability, proper functioning, and intended use.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the corrosion-resistant steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. Unit of product for examination purposes shall be one tibial bolt.

Table II. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. Unit of product for test purposes shall be one tibial bolt. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table III. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Insp. level	AQL
Tibial bolt (all components)	Grain size	3.1.1	ASTM F55	- - -	- -
	Inclusion content	3.1.2	ASTM E45	S-1*	1.0
	Hardness	3.2	4.4.2	S-2	1.5
	Dimensions	3.4	- - - -	S-2	1.5
	External thread test of bolt	3.5	4.4.3	S-2	1.0
	Internal thread test of nuts	3.6	4.4.4	S-2	1.0
	Boil test	3.7	4.4.1.1	S-4	0.65
	Copper sulfate test	3.7	4.4.1.2	S-4	0.65
	Saline test	3.7	4.4.1.3	S-4	0.65

* The unit lot of steel, for which the inclusion content is to be determined, shall not be larger than one heat.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the tibial bolts shall be selected for chemical analysis.

4.3 Examination. The tibial bolts shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any bolts in the sample containing one or more defects shall be rejected and if the number of defective bolts in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

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Table IV. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	One or more components missing.
102	Bolt head does not possess screwdriver slot.
103	Threads of bolt not complete or well defined.
104	All components not free from burrs, pits, cracks, slivers, or tool marks.
105	Components not free from scale, corrosion, or foreign matter.
106	Point of bolt not sharp.
107	Finish of components not as specified.
108	Nuts cannot be easily turned on bolt threads by hand.
109	Washers do not possess sufficient clearance hole for bolt threads or bolt shoulder.
Minor	
201	Bolts not properly protected in unit package.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. All components of the tibial bolt shall be subjected to the boil test as described and specified in Federal Specification GG-I-526.

4.4.1.2 Copper sulfate test. All components of the tibial bolt shall be subjected to the copper sulfate test as described and specified in GG-I-526.

4.4.1.3 Saline test. All components of the tibial bolt shall be scrubbed with soap and warm water, rinsed in hot water, followed by immersion into ethyl alcohol (95 percent) and dried. The components shall then be completely immersed in Normal Saline Solution (U.S.P.) and while immersed, placed into an oven at a temperature of 37 $\pm$ 1 degrees Centigrade for a period of 7 days, removed from the oven and examined for corrosion while still immersed.

4.4.2 Hardness test. Rockwell hardness of the tibial bolt shall be determined in accordance with method 243 of Federal Test Method Standard No. 151.

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4.4.3 External thread test. The contractor shall provide the necessary thread ring gages to the Government Quality Assurance Representative (QAR) for checking the external threads of the bolt in accordance with the required thread dimensions. The following dimensional notations shall be inscribed on the ring gages:

Thread Size:	3-48 UNC-2A
"GO" PITCH DIAMETER:	0.0848
"LO" PITCH DIAMETER:	0.0825
"GO" MAJOR DIAMETER:	0.0983
"NOT GO" MAJOR DIAMETER:	0.0938

The gaging procedure to be used, dimensions, tolerances and gage identifications shall be in accordance with A.N.S.I. Standard B1.2. Fits of gages onto the bolt threads shall be noted.

4.4.4 Internal thread test. The contractor shall provide the necessary thread plug gages to the QAR for checking the internal threads of the nuts in accordance with the required thread dimensions. The following dimensional notations shall be inscribed on the plug gages:

Thread Size:	3-48 UNC-2B
"GO" PITCH DIAMETER:	0.0855
"HI" PITCH DIAMETER:	0.0885
"GO" MINOR DIAMETER:	0.0764
"NOT GO" MINOR DIAMETER:	0.0845

The gaging procedure to be used, dimensions, tolerances and gage identifications shall be in accordance with A.N.S.I. Standard B1.2. Fits of gages into nuts shall be noted.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One bolt along with 2 nuts and 2 washers, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	480 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-1.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in Table II of PPP-B-636 shall apply for fiberboard boxes.

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5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.2 This specification is intended to cover Federal Stock Number 6515-310-5000 BOLT, TIBIAL, appearing in the Federal Supply Catalog. It does not include all types as indicated by the title nor all types commercially available.

6.3 Supersession data. This specification includes requirements of Purchase Description No. 5, dated 15 March 1971.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1586

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be used on the information in the current DODISS.

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